Opt on manifold Notes

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Chapter 1

Optimization theory on linear spaces

These are brief notes based on chapters 5 of [8] and chapter 4 and 5 of [4]. We consider optimization problems for functions defined on subsets of linear spaces. Some proofs are presented in a geometric manner. The section on duality refers to chapter 4 of [3] to illustrate an interesting relation between Fenchel duality and Lagrangian duality. For simplicity, we only consider proper functions: those taking values in $(-\infty, \infty]$ and not identically infinity.

1.1 Existence and Uniqueness

It's well-known that if f is lower semicontinuous and domain is compact, then f has a minimum: suppose its codomain does not have a minimal element, then the open cover

$$\{f^{-1}(f(x), \infty]\}$$

has finite subcover.[1] Thus if a closed sublevel set $C = \{f \leq c\}$ is bounded and nonempty for some $c \in \mathbb{R}$, then f has a minimum over C hence has a global minimum. A typical example is when f is coercive, i.e. $\lim_{|x|\to\infty} f(x) = \infty$, then the sublevel sets are bounded and nonempty for sufficiently large c.

On the other hand, if f is lower semicontinuous and strictly quasiconvex(for any segment connecting points of graph of f, the graph lies strictly below maximum of the endpoints; this implies the sublevel sets are strictly convex), then such minimum must be unique.

[1] Another way to put this: the funciton is lower semicontinuous iff it's continuous into $\bar{\mathbb{R}}$ under right order topology, whose compact set has a minimal element

1.2 Optimization condition of unconstrained optimization

The smooth case is well-known: if f is differentiable, then a necessary condition for x to be a local minimum point is $\nabla f(x) = 0$ and $\mathrm{Hess} f(x) \succeq 0$ if f is twice-differentiable, and a sufficient condition is $\nabla f(x) = 0$ and $\mathrm{Hess} f(x)$ is positive-definite. These conditions also applied to functions on manifolds.

If the function is convex, then the necessary and sufficient condition for x to be a global minimum point is $0 \in \partial f(x)$. This is necessary even if f is not convex, with the definition of subdifferential suitably generalized.

1.3 Duality theory

From now on we consider the optimization problem with constraints, the duality theory is very useful to understand the optimality condition of such problems. The idea is to convert the problem to dual problem with better constraints and computable optimization condition.

1.3.1 Fenchel duality

So basically we are extending Legendre transform to general cases: the Frenchel dual of f is defined as

$$f^*(y) = \sup_x \langle x, y\rangle - f(x)$$

It can be understood as the restriction of the support function of the epigraph of f:

$$f^*(y) = \sup_{(x,t)\in \mathrm{epi}(f)} \langle (x,t), (y,-1)\rangle$$

which detects the shape of epigraph f by its supporting hyperplanes (although this only record information of its convex hull). This dual function is always convex and lower-semicontinuous as they are of a supremum of affine functions, and we have the Fenchel inequality:

$$f(x) + f^*(y) \geqslant \langle x, y\rangle$$

which immediately implies that the bidual $f^{**} \leq f$. If f is convex and lower-semicontinuous, then we have $f^{**} = f$. This is easy if we assume subdifferential exists, in which case we have nice summetry for points (x, y) that equalizing the Frenchel inequality: $x \in \partial f^*(y), y \in \partial f(x)$. Also, $*$ is order reversing, so in general, f^{**} is the largest lower-semicontinuous convex function that is less than or equal to f, i.e. epigraph of f^{**} is the convex hull of epigraph of f.

When function is smooth, f^* is just extension of Legendre transform. For example if $f(x) = e^x$ then

$$f^*(y) = \sup_x xy - e^x = \begin{cases} y\log y - y & y > 0 \\ 0 & y = 0 \\ \infty & y < 0 \end{cases}$$

An important non-smooth example is the indicator function on some set C:

$$I_C = \begin{cases} 0 & x \in C \\ \infty & x \notin C \end{cases}$$

we have

$$I_C{}^*(y) = \sup_{x \in C} \langle x, y \rangle - I_C(x) = \sup_{x \in C} \langle x, y \rangle$$

which is the support function of C. For a convex cone C, the dual of the support function is again a support function:

$$I_C^* = I_{C^0}$$

where

$$C^0 = \{y : \langle x, y \rangle \leq 0 \text{ for all } x \in C\}$$

is the polar cone of C.

1.3.2 Perturbation

This section refers to chapter VI of [5]. We consider a perturbation $F(x, \lambda)$ of f such that $F(x, 0) = f(x)$ and a family of optimization problem $p(\lambda) = \inf_x F(x, \lambda)$, then

$$p^*(\lambda) = \sup_s \left(\langle \lambda, s \rangle - \inf_y F(y, s) \right) = \sup_{s,y} \left(\langle \lambda, s \rangle - F(y, s) \right) = F^*(0, \lambda)$$

The primal program is $p(0) = \inf_x F(x, 0)$, the dual program is

$$p^{**}(0) = \sup_\lambda \left(\langle 0, \lambda \rangle - u^*(\lambda) \right) = \sup_\lambda \left(-F^*(0, \lambda) \right) = -\inf_\lambda F^*(0, \lambda)$$

This can be viewed as the promgram associated with F^*. If the bidual $F^{**} = F$, then $(P^{**}) = (P)$.

Henceforth, we omit the subscripts of inf and sup, understanding that $\inf F(\lambda) := \inf_x F(x, \lambda)$ and $\inf F^*(x) := \inf_\lambda F(x, \lambda)$. For example we shall write $p = \inf F$ and $p^{**} = -\inf F^*$. We have weak duality

$$\sup \left(-F^* \right)(0) \leq \inf F(0)$$

since $p^{**}(0) \leq p(0)$ and

$$p(0) - p^{**}(0) = \sup \left(-F^* \right) - \inf F$$

is the duality gap. If the gap is 0(this happens if p is convex and lower semicontinuous at 0), we say strong duality holds.

If F is convex, then $p(\lambda) = \inf_{x \in C} F(x, \lambda)$ is always convex as long as $p > -\infty$. This is because the epigraph of p is the projection of the epigraph of F, and the projection of a convex set is always convex.

The relative interior $\mathbf{Relint}(D)$ of a set D is the interior of D relative to its affine hull. Convex functions are continuous on the relative interior of their domain, so if $0 \in \mathbf{Relint}\,(\mathrm{dom}(p))$, then p is continuous at 0. In this case, the strong duality holds since p is continuous near 0. In this case, we say the convex program is strongly consistent, so basically we have a nice perturbation of f.

1.3.3 Lagrangian duality

It is heuristic to consider the smooth case first. Suppose we have a constrained optimization problem

$$\min_{x \in C} f(x) \text{ s.t. } g(x) = 0$$

If g has constant rank, then the first order condition is that df annihilates the tangent space Kerdg, this happens iff $df = \langle \lambda, dg \rangle$ dualizing via a metric, we see

$$\nabla f(x) = \lambda \nabla g(x)$$

It has exactly the same first order optimization condition as the unconstrained minimizing problem of Lagrangian function

$$L(x, \lambda) = f(x) + \lambda g(x)$$

A constrained optimization problem is

$$\inf_x f(x) \text{ s.t. } g(x) \geq_K 0 \tag{P}$$

where K is a convex cone, for example a hemispace $H = [-\infty, 0]^p \times \{0\}^q$ We can convert the problem to an unconstrained optimization for

$$F(x) = f(x) + I_K(g(x))$$

where I_K is the indicator function on K.

Let's apply the insight of the last section to F. We consider the perturbation

$$F(x, \lambda) = f(x) + I_K(g(x) + \lambda)$$

Then

$$\begin{aligned}
-F^*(0, \lambda) &= \inf_{y,s} \left(-\langle \lambda, s \rangle + f(y) + I_K(g(y) + s)\right) \\
&= \inf_{y,r} \left(-\langle \lambda, r - g(y) \rangle + f(y) + I_K(r)\right) \\
&= \inf_{y} \left(f(y) + \langle \lambda, g(y) \rangle\right) - I_K^*(\lambda) \\
&= \inf_{y} \left(f(y) + \langle \lambda, g(y) \rangle\right) - I_{K^0}(\lambda)
\end{aligned}$$

Consider the Lagrangian:

$$L(y, \lambda) = f(y) + \langle \lambda, g(y) \rangle$$

with $\lambda \in K^0$ in mind. It can be under stood as an approximation of the original function $F(x) = f(x) + I_K(g(x))$ by a family of linear functions. We call

$$L(\lambda) = \inf_y L(y, \lambda)$$

the Lagrangian dual function.

Now we can write

$$-p^*(\lambda) = -F^*(0,\lambda) = \inf_y L(y,\lambda) - I_{K^0}(\lambda) = L(y) - I_{K^0}(\lambda)$$

The dual program of P is to minimize p^*, it can be written as

$$\sup_\lambda (L(\lambda)) \text{ s.t. } \lambda \leq_{K^*} 0 \tag{P^*}$$

In the hemispace case, the polar cone and dual cone is

$$H^0 = [0,\infty]^p \times \bar{\mathbb{R}}^q, H^* = -H^0 = [-\infty,0]^p \times \bar{\mathbb{R}}^q$$

One can also interpret the Lagrangian dual from the perspective of minimax theory:

$$\sup_{\lambda\in K^0} L(\lambda) = \sup_\lambda \inf_y (L(y,\lambda) - I_{K^0}(\lambda)) \leq \inf_y \sup_\lambda (L(y,\lambda) - I_{K^0}(\lambda)) = \inf_{g(y)\in K} f(y)$$

with equality at a saddle point of $L(y,\lambda), \lambda \in K^0$, in this case strong duality holds.

We shall see later that for convex problems, a saddle point will always be an optimal vector, the converse also true if Slater's condition holds.

There is a more genral duality theorem[5]: one have

$$\inf(f-g) = \sup(g_* - f^*)$$

when $f, g, f-g$ are proper convex. TBA

1.4 Optimization condition for general constraints

1.4.1 Tangent cone and constraint qualifications

The tangent cone $\mathscr{T}(x)$ (in the sense of the paratingent cone or contingent cone of Bouligand) at a feasible point x consists of all directions d such that there is a feasible sequence $x_k \to x$ and positive numbers t_k s.t.

$$(x_k - x)/t_k \to d$$

This should be equivalent to

$$\mathscr{T}(x) := \left\{ d : \liminf_{t\to 0+} \frac{d(x+td, C)}{t} = 0 \right\}$$

If the feasible set is convex, $\mathscr{T}(x)$ is the closure of the cone consists of open rays that intersect nontrivially the feasible set.

Cones of feasible directions defined above are not that computable. We need to introduce following linear approximations of $\mathscr{T}(x)$. Geometrically, these are

$$\mathfrak{T}(x) = (dF)^{-1}(\mathscr{T}_{g(x)}K)$$

where g is the constraint function and $\mathscr{T}_{g(x)}K$ is the tangent cone of the constraint set at $g(x)$. In general, we have $\mathscr{T}(x) \subset \mathfrak{T}(x)$, and the equality holds if the constraint is regular at x.

For simplicity, we restricted ourself to the case when the cone K is the simpliest manifold with corner H from now on, i.e. we are considering the problem

$$\inf_x f(x) \text{ s.t. } g(x) \leq 0, h(x) = 0$$

At a fixed feasible point x, we rearrange the constraints $g = (g_1, g_2)$ s.t. $g_1(x) = 0, g_2(x) < 0$, so we say g_1, h are active constraints, and g_2 is inactive in a neighborhood of x. The necessary condition for a feasible direction is

$$\mathfrak{T}(x) = \{d : \langle \nabla h, d\rangle = 0, \langle \nabla g_1, d\rangle \leq 0\}$$

This might be bigger than $\mathscr{T}(x)$, for example the linearized cone of feasible directions of the constraints $g_1(x) = x^3 \leq 0$ or $h(x) = x^3 = 0$ will give false feasible direction at 0. But if g_1, h is regular at x (namely its differential has full rank), then the level set $g_1 = 0, g_2 \leqslant 0$ defines a manifold with corner near x. In this case $\mathscr{T}(x) = \mathfrak{T}(x)$. We say Linear independence constraint qualification(**LICQ**) holds at x if the gradients of active constraints are linearly independent.

We actually do not need $(\nabla g_1, \nabla h)$ to have full rank, constant rank theorem tells us that it suffices the rank of each subset of the gradients of the active constraints at a vicinity of x is constant(**CRCQ**). In this case, for a fixed $d \in \mathfrak{T}(x)$, the active constraints with vanishing gradient along d will contains a submanifold that tangent to d, thus we can choose a curve along d that lies within such submanifold, while the rest of the active constraints with non-vanishing gradient along d will become inactive along such curve. A special case of these is linear constraints qualification(**LCQ**), when g_1, h are affine functions.

We also say Mangasarian-Fromovitz constraint qualification (**MFCQ**) holds at x if the gradients ∇h of equality constraints are linearly independent and there is a direction d_0 s.t. $\langle \nabla h, d_0\rangle = 0, \langle \nabla g_1, d_0\rangle < 0$ at x. In this case we choose curves $x(t, s)$ along $d + sd_0$ that lies within the submanifold defined by $h = 0$, in orther words

$$x(t, s) - x(0, 0) = (d + sd_0)(t + o(t))$$

now $x(t, t)$ will be a curve in the feasible set along d.

1.4.2 First order condition: Karush-Kubn-Tucker condition

At a local minima, either the function doesn't descent, or it meets the boundary of the feasible set. Thus if our linearized tangent cone is nice:

$$\mathscr{T}(x) = \mathfrak{T}(x)$$

then the following

$$\langle d, \nabla f\rangle < 0, \langle d, \nabla h\rangle = 0, \langle d, \nabla g_1\rangle \leq 0$$

cannot be true at the same time at x.

Farkas lemma says these conditions are equivalent to the existence of λ and nonnegative μ s.t.

$$-\nabla f = \lambda \cdot \nabla h + \mu \cdot \nabla g_1$$

(here I use $\cdot$ to distinguish the product w.r.t. the index of constraints with the inner product on space) To see this, consider the convex cone

$$C = \{\lambda \cdot \nabla h + \mu \cdot \nabla g_1 : \mu \geq 0\}$$

the second and third conditions are equivalent to $-d \in C^*$, for any such d, $\langle d, \nabla f \rangle \geq 0$ thus $-\nabla f$ must lies in C^{**}. But the bidual of a closed convex cone is itself(by the hyperplane separation theorem), so $-\nabla f \in C$.

Now we understand that a necessary condition for a local minima x is the existence of λ and nonnegative μ s.t. satisfies $\partial_x L(x, \lambda, \mu) = 0$. We can put these conditions together with the information of active constraints: the **KKT condition** is

Stationarity:	$\partial_x L(x, \lambda, \mu) = \nabla f(x) + \lambda \cdot \nabla h(x) + \mu \cdot \nabla g(x) = 0$
Primal feasibility:	$g(x) \leq 0,\ h(x) = 0$
Dual feasibility:	$\mu \geq 0$
Complementary slackness:	$\mu \cdot g(x) = 0$

(λ, μ) are called the KKT multipliers. The KKT condition is necessary for local optimality if the linearized cone of feasible directions is nice, which amounts to verify various constraint qualifications.

If we interpret f as a potential field, the g, h as (one-sided or two-sided) constaints, and the optimal problem as solving the equilibrium of the system, then the KKT condition is just the balance of forces: the potential force ∇f is balanced by the constraint forces $\lambda \cdot \nabla h + \mu \cdot \nabla g$. Of cource we need ∇g and ∇h to be nice enough to be able to stands for the normal forces of the constraints, which is exactly the constraint qualification condition.

The KKT condition is also natural in terms of the Lagrangian duality: it's first order condition of the critical point of the Lagrangian function. Thus, KKT conditions are necessary if the strong duality holds. TBA

1.4.3 Second order condition

The cone of critical directions is defined as

$$\mathscr{C}(x) = \{d \in \mathfrak{T}, \langle d, \nabla g_1 \rangle = 0\}$$

A second order necessary condition for local optimality at a KKT point (x, λ, μ) is that $\mathrm{Hess}_x L(x, \lambda, \mu)$ is positive semidefinite on $\mathscr{C}(x)$. A second order sufficient condition for local optimality is that $\mathrm{Hess} L(x, \lambda, \mu)$ is positive definite on $\mathscr{C}(x)$.

1.5 Optimization condition for convex problems

1.5.1 Slater's qualification and strong duality

Now we consider the problem P with convex objective function and affine equality and convex inequality constraints:

$$\min_{x \in D} f(x) \text{ s.t. } g(x) \leq 0, Ax = b$$

where f is proper and convex, g is convex and A is a linear map. As before, for convex problems we do not require the function to be differentiable. The perturbation of this problem is

$$F(x, \lambda, \mu) = f(x) + I_H((g(x), Ax - b) + (\lambda, \mu))$$

where $H = [-\infty, 0]^p \times \bar{\mathbb{R}}^q$ is the hemispace. The Lagrangian function is

$$L(x, \lambda, \mu) = f(x) + \lambda \cdot (Ax - b) + \mu \cdot g(x)$$

and

$$p(\lambda, \mu) = \inf_x (f(x) + I_H((g(x), Ax - b) + (\lambda, \mu)))$$

and dual function

$$p^*(\lambda, \mu) = -L(\lambda, \mu) + I_{H^0}(\lambda, \mu)$$

Recall our dicussion on perturbation and strong duality in 1.3.2. The condition $0 \in \mathbf{Relint}\,(\mathrm{dom}(p))$ is equivalent to the following condition: there is a point x s.t.

$$g_0(x) = 0, g_1(x) < 0, Ax = b$$

where g_0 is the affine part of g. This is the **Relaxed Slater's qualification** for the problem P. A special case is when $x \in \mathbf{Int}(\mathrm{dom}(p))$, this is equivalent to **Slater's qualification**

$$g(x) < 0, Ax = b$$

In these cases, the strong duality holds since p is continuous at 0.

1.5.2 Convex KKT

When Slater's condition holds, the strong duality holds, we have seen that if the functions are differentiable, the KKT condition is necessary for optimality. Note that MFCQ is a special case of this.

The convex case is slightly different. The KKT condition for convex problems is the same as before but with the differential replaced by subdifferential:

Stationarity:	$0 \in \partial_x L(x, \lambda, \mu) = \partial f(x) + \lambda \cdot A + \mu \cdot \partial g(x)$
Primal feasibility:	$g(x) \leq 0,\ Ax = b$
Dual feasibility:	$\mu \geq 0$
Complementary slackness:	$\mu \cdot g(x) = 0$

Convex KKT is sufficient for strong duality and optimazation. At the KKT point $L(x, \lambda, \mu) = f(x) \geqslant \inf f$, which imlies the duality gap is zero and the solution is optimal.

Under Slater's condition, the convex KKT condition is also necessary for optimality, but the proof should involve subdifferentials (TBA).

Chapter 2

Algorithms

I briefly summarize the algorithms for optimization problems in this chapter.

2.1 Gradient descent

2.1.1 Exact line search

We assume throughout the section that f is diffenrentiable and attains a minimum $\inf f = f(x^\star)$. The idea of GD is simple:

$$x_{k+1} = x_k + t_k d_k$$

where $d_k = -\nabla f(x_k)$ is a descent direction t_k is the step size.

The real problem is how to choose the step size. You can use **exact line search** to find the optimal step size at each step. i.e.

$$t_k = \arg\min_{t>0} f(x_k + t d_k)$$

This implies that

$$\langle \nabla f(x_{k+1}), d_k \rangle = \langle \nabla f(x_{k+1}), -t_k \nabla f(x_k) \rangle = 0$$

So the descent direction in GD is orthogonal to last step and sequence will zigzag along a polygonal curve. A very important example:

2.1.1 Example (Quadratic form)

If we have a positive definite quadratic form

$$f(x) = \frac{1}{2} \langle x|A|x \rangle = \frac{1}{2}|x|_A^2$$

then the optimal direction is

$$d_k = -Ax_k$$

step size is

$$t_k = \frac{|Ax_k|^2}{|Ax_k|_A^2} \subset [1/\sigma_1, 1/\sigma_r]$$

thus

$$\begin{aligned}|x_{k+1}|_A^2 =& |x_k - t_k A x_k|_A^2 = |x_k|_A^2 + t_k^2 |Ax_k|_A^2 - 2t_k \langle x_k|A|x_k\rangle_A \\ =& \left(1 - \frac{|Ax_k|^4}{\langle Ax_k|A|Ax_k\rangle \langle x_k|A|x_k\rangle}\right)|x_k|_A^2 \\ =& \left(1 - \frac{|d_k|^4}{\langle d_k|A|d_k\rangle \langle d_k|A^{-1}|d_k\rangle}\right)|x_k|_A^2\end{aligned}$$

by Cauchy-Schwarz inequality (or in this case its called **Kantorvich inequality**) for point

$$(a, b) = \left(\sum \sigma_j y_j^2, \sum_i \frac{y_i^2}{\sigma_i}\right)$$

in the convex hull of the set $\{(\sigma_i, 1/\sigma_i) : i = 1, \cdots, r\}$, the maximal value of $a \times b$ lies on the middle point of the segment connecting $(\sigma_1, 1/\sigma_1)$ and $(\sigma_r, 1/\sigma_r)$, which is

$$b = -\frac{1}{\sigma_1\sigma_r}a + \frac{\sigma_1 + \sigma_r}{\sigma_1\sigma_r}$$

thus

$$\left(\sum_i \frac{y_i^2}{\sigma_i}\right)\left(\sum \sigma_j y_j^2\right) \leq \frac{\sigma_1 + \sigma_r}{2\sigma_1\sigma_r}\sum y_i^2$$

As a consequence, we have

$$|x_{k+1}|_A^2 \leq \left(1 - \left(\frac{2\sigma_1\sigma_2}{\sigma_1 + \sigma_2}\right)^2\right)|x_k|^2 = \left(\frac{\sigma_1 - \sigma_r}{\sigma_1 + \sigma_r}\right)^2 |x_k|_A^2$$

so the sequence x_k has a linear convergence rate with factor

$$\left(\frac{\kappa - 1}{\kappa + 1}\right)^2$$

where κ is the condition number of A. When $\kappa = \frac{\sigma_1}{\sigma_r}$ is large, the shape of the graph of f looks like a narrow valley, the convergence is very slow as the sequence gets zigzag in the valley.

We shall generalise these observations to functions with bounded Hessians, with σ_1 control the convexity and σ_r control the smoothness of f.

If f is proper, L-smooth

$$f(y) \leq f(x) + \langle \nabla f(x), y - x\rangle + \frac{L}{2}\left|y - x\right|^2$$

If we are using the exact line search

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L}\left|\nabla f(x_k)\right|^2$$

If f is also μ convex,

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2}|y - x|^2 \geq f(x) - \frac{1}{2\mu}|\nabla f|^2$$

then

$$\kappa \left(f(x_k) - f(x_{k+1}) \right) \geq f(x_k) - \inf f$$

where $\kappa = \frac{L}{\mu}$ is the condition number of $\nabla^2 f$. It follows that

$$f(x_{k+1}) - \inf f \leq \frac{\kappa - 1}{\kappa} \left(f(x_k) - \inf f \right)$$

which means the convergence is linear with factor

$$1 - \frac{\mu}{L} = \frac{\kappa - 1}{\kappa}$$

which is close to 1 for large κ.

The bound rate of convergence we derived here is bigger than the quadratic case, but in fact it was proved in 2016 by Klerk et-al [6] that the bound of the rate of convergence of exact line search is actually

$$\left(\frac{L - \mu}{L + \mu} \right)^2 = \left(\frac{\kappa - 1}{\kappa + 1} \right)^2$$

for the worst case.(TBA)

2.1.2 Back-track line search

In general, we use inexact line search methods. But first lets look at the case when we have a good guess of the Lipschitz constant L of ∇f.

Lemma 2.1.2 (Descent lemma of objective values)

If f is convex and L-smooth, $t_k \in [t_-, t_+] \subset (0, 1/L]$:

$$f(x_{k+1}) \leq f(x^\star) + \frac{1}{2t_k}|x_k - x^\star|^2 - \frac{1}{2t_k}|x_{k+1} - x^\star|^2$$

Proof.

$$\begin{aligned}
f(x_{k+1}) \leq & f(x_k) - t_k|\nabla f(x_k)|^2 + \frac{L}{2}t_k^2|\nabla f(x_k)|^2 \\
\leq & f(x^\star) + \langle \nabla f(x_k), x_k - x^\star \rangle - \frac{1}{2}t_k|\nabla f(x_k)|^2 \\
= & f(x^\star) + \left\langle \nabla f(x_k), x - x^\star - \frac{1}{2}t_k \nabla f(x_k) \right\rangle \\
= & f(x^\star) + \frac{1}{2t_k}|x_k - x^\star|^2 - \frac{1}{2t_k}|x_{k+1} - x^\star|^2
\end{aligned}$$

□

As a consequence

$$(k+1)t_-(f(x_k)-f(x^\star)) \le \sum_k t_k(f(x_{k+1})-f(x^\star)) \le \frac{1}{2}|x_0-x^\star|^2$$

and

$$f(x_k)-f^\star \le \frac{|x_0-x^\star|^2}{2kt_-}$$

which implies the convergence rate is $O(1/k)$.

As in the exact line search case, we have

Theorem 2.1.3

If f is μ-convex and L-smooth, the stepsize are chosen so that

$$t_k \in [t_-, t_+] \subset (0, 2/\mu + L]$$

we have linear convergence rate:

$$1 - t_- \frac{2\mu L}{\mu + L} \ge \left(\frac{L-\mu}{L+\mu}\right)^2.$$

Again note that the convergence rate has the same bound as exact line search.

Proof. We use the inequality in B.1.5:

$$\langle \nabla f(x) - \nabla f(y), x-y\rangle \ge \frac{\mu L}{\mu+L}|x-y|^2 + \frac{1}{\mu+L}|\nabla f(x)-\nabla f(y)|^2$$

to get

$$\begin{aligned}|x_{k+1}-x^\star|^2 =& |x_k-x^\star|^2 - 2t_k\langle \nabla f(x_k), x_k - x^\star\rangle + t_k^2|\nabla f(x_k)|^2 \\ \le& \left(1 - 2t_k\frac{\mu L}{\mu+L}\right)|x_k-x^\star|^2\end{aligned}$$

Also using

$$f(x_{k+1})-f(x^\star) \le \frac{1}{2t_k}\left(|x_k-x^\star|^2 - |x_{k+1}-x^\star|^2\right) \le \frac{1}{2t_k}\left(1 - t_k\frac{2\mu L}{\mu+L}\right)^k |x_0-x^\star|^2$$

□

TBA

But the problem is most of the time we don't know L beforehand. To determine t_k, we use the **back-track line search**. You start with a stepsize then reduce it by a factor $\tau \in (0,1)$ until the **Armijo condition** is satisfied

$$f(x_{k+1}) - f(x_k) \le rt_k\langle \nabla f(x_k), d_k\rangle$$

where $r \in (0,1)$ is understood as the ratio of the real descent and expected descent. We also assume our function is L-smooth:

$$f(x_{k+1}) - f(x_k) \le t_k\langle \nabla f(x_k), d_k\rangle + \frac{L}{2}t_k^2|d_k|^2$$

In case of $d_k = -\nabla f(x_k)$, we have

$$-t_k|\nabla f(x_k)|^2 + \frac{L}{2}t_k^2|\nabla f(x_k)|^2 < -rt_k|\nabla f(x_k)|^2$$

once

$$t_k < \frac{2(1-r)}{L}$$

so the output of the backtrack line search is at most $\frac{2(1-r)}{L}\tau$ and the number of iterations we need is $\sim \log_\tau\left(\frac{1-r}{L}\right)$.

2.1.3 Barzilai-Borwein

We have seen in the last section that the convergence of gradient descent is very slow when the condition number κ is large. The Barzilai-Borwein method is a simple modification of gradient descent to improve the convergence in this case. Let $\Delta x_k = x_k - x_{k-1}, \Delta g_k = \nabla f(x_k) - \nabla f(x_{k-1})$. The **long BB step**:

$$\alpha_k^L = \frac{|\Delta x_k|^2}{\langle \Delta x_k, \Delta g_k\rangle}$$

and the **short BB step**:

$$\alpha_k^S = \frac{\langle \Delta x_k, \Delta g_k\rangle}{|\Delta g_k|^2}$$

Note that if f is μ-convex and L-smooth,

$$\frac{1}{L} \le \alpha_k^S \le \alpha_k^L \le \frac{1}{\mu}$$

The stepsize can also be viewed as the best approximation of inverse of Hessian which satisfies the following intermediate value equality(or secant equation):

$$\Delta x_k - \left(\nabla^2 f(\xi)\right)^{-1}\Delta g_k = 0, \xi \in [x_{k-1}, x_k]$$

α are chosen to minimize

$$|\Delta x_k/\alpha - \Delta g_k|$$

or

$$|\Delta x_k - \alpha\Delta g_k|^2$$

In orther words, BB method is actually a simplified quasi-Newton method: it's just the secant methodalong the search direciton.

Also this method is non-monotone. Thus we need to shrink α until the following Grippo's condition is satisfied

$$f(x_k + \alpha d_k) \le \max_{j\in[\min\{k,M\}]} f(x_{k-j}) + r\alpha\langle \nabla f(x_k), d_k\rangle$$

so that f doesn't increase too much compared to the preceding M steps.

2.2 Newton's method

2.2.1 Damped Newton's method

We have seen in the last section that the convergence of gradient descent is very slow when the condition number κ is large, so it performs poorly if you choose bad coordinates. We can either modified the stepsize as the BB method, or change the search directions.

Recall in the example 2.1.1, the gradient descent use the search direction $d_k = -\nabla f(x_k) = -Ax$, but if we are in god mode and know the Hessian A, we should have used the optimal direction $x^* - x = -x$ with stepsize 1 which is $-A^{-1}\nabla f$. Thus the **classical or pure Newton's method** is defined as the iteration of the following update rule:

$$x_{k+1} = x_k + d_k \text{ s.t. } \nabla^2 f(x_k) d_k = \nabla f(x_k)$$

which minimize the second order approximation of f at each step. The Newton step (if Hessian is non-singular)

$$-(\nabla^2 f(x_k))^{-1} \nabla f(x_k)$$

is invariant under the affine transformation of coordinates, so it is not heavily influenced by the condition number as in GD.

The **Newton decrement** is

$$\lambda_k^2 = -\frac{d}{dt} f(x + td_k) = \langle -\nabla f(x_k), d_k \rangle = \left\langle \nabla f(x_k) | \nabla^2 f(x_k)^{-1} | \nabla f(x_k) \right\rangle$$

can be thought as an estimate of $f - \inf f$. When λ_k is small, we need to shrink stepsize using backtrack line search. TBA

2.2.2 Quasi-Newton method

Sometimes computing Hessian is costly, we use the information of succesive gradients to approximate the Hessian which satisfies the following secant equation

$$\nabla^2 f(\xi) \Delta x_k - \Delta g_k = 0, \xi \in [x_{k-1}, x_k]$$

Let $\Delta x_k = x_k - x_{k-1}, \Delta g_k = \nabla f(x_k) - \nabla f(x_{k-1})$. We choose B_k or its inverse H_k that satisfying following **secant equation**

$$\Delta g_k = B_k \Delta x_k$$

or inversely

$$H_k \Delta g_k = \Delta x_k$$

The iteration rule is then

$$x_{k+1} = x_k - \alpha_k B_k^{-1} \nabla f(x_k) = x_k - \alpha_k H_k \nabla f(x_k)$$

You absolutely want $B_k \succeq 0$ so at least

$$\langle \Delta g_k, \Delta x_k \rangle > 0$$

This is called the **curvature condition**. It is guarenteed by backtrack line search using Wolfe condition

$$\langle \nabla f(x_k + \alpha d_k), d_k \rangle \geq (1 - c) \langle \nabla f(x_k), d_k \rangle$$

The simpliest update formular of B is the following **symmetric rank-one (SR1) method**:

$$B_k = B_{k-1} \pm uu^*$$

so

$$\pm u \langle u, \Delta x_k \rangle = \Delta g_k - B_{k-1} \Delta x_k$$

thus

$$u = \frac{\Delta g_k - B_{k-1} \Delta x_k}{\sqrt{\pm \langle \Delta x_k, \Delta g_k - B_{k-1} \Delta x_k \rangle}}$$

and

$$B_k = B_{k-1} + \frac{(\Delta g_k - B_{k-1} \Delta x_k)(\Delta g_k - B_{k-1} \Delta x_k)^*}{\langle \Delta x_k, \Delta g_k - B_{k-1} \Delta x_k \rangle}$$

And we can invert B_k via the Sherman-Morrisan formulaA.2.1

$$H_k = H_{k-1} + \frac{(\Delta x_k - H_{k-1} \Delta g_k)(\Delta x_k - H_{k-1} \Delta g_k)^*}{\langle \Delta x_k, \Delta x_k - H_{k-1} \Delta g_k \rangle}$$

Note the formula requires the denominator

$$\langle \Delta g_k - B_{k-1} \Delta x_k, \Delta x_k \rangle \neq 0$$

Also note that we positive-definiteness is not preserved if the denominator is negative.

There is also rank two updates: the **Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm** is formula of the form:

$$B_k = B_{k-1} + auu^* + bvv^*$$

which implies

$$a \langle u, \Delta x_k \rangle u + b \langle v, \Delta x_k \rangle v = \Delta g_k - B_{k-1} \Delta x_k$$

An obvious solution is

$$u = \frac{\Delta g_k}{|\langle \Delta g_k, \Delta x_k \rangle|^{1/2}}, v = \frac{B_{k-1} \Delta x_k}{\langle \Delta x_k, B_{k-1}, \Delta x_k \rangle^{1/2}}$$

Thus the updating rule is

$$B_k = B_{k-1} + \frac{(\Delta g_k)(\Delta g_k)^*}{\langle \Delta g_k, \Delta x_k \rangle} - \frac{(B_{k-1} \Delta x_k)(B_{k-1} \Delta x_k)^*}{\langle \Delta x_k | B_{k-1} | \Delta x_k \rangle}$$

Again using the SM formula to invert B_k we have

$$H_k = \left(1 - \frac{(\Delta x_k)(\Delta g_k)^*}{\langle \Delta g_k | \Delta x_k \rangle}\right) H_{k-1} \left(1 - \frac{(\Delta g_k)\,(\Delta x_k)^*}{\langle \Delta g_k | \Delta x_k \rangle}\right) + \frac{|\Delta x_k|^2}{\langle \Delta g_k | \Delta x_k \rangle}$$

Note that if the curvature condition is satisfied, then the formula preserves the positive-definiteness of H_{k-1}.

We can also use rank two updates on H_k, this yields the following **Davidon-Fletcher-Powell (DFP) formula**:

L-BFGS

2.3 Trust-region method

The idea is we choose step size first, then we determine the search direction (in contrast with the line search method). At each iteration we minimize a quadratic approxiamation

$$m_k(d) = f(x_k) + \langle \nabla f(x_k), d \rangle + \frac{1}{2} \langle d | B_k | d \rangle$$

of the objective function at x_k in a ball (the **trust-region**) with radius Δ_k

$$\min m_k(d), |d| \leq \Delta_k$$

The radius is updated according to the , ratio of actual reduction and expected reduction:

$$\rho_k = \frac{f(x_k) - f(x_k + d_k)}{m_k(0) - m_k(d_k)}$$

2.3.1 Trust-region subproblem

The TRS is of the form:

$$\min m(d) = f + \langle g, d \rangle + \frac{1}{2} \langle d | B | d \rangle \,, |d| \leq \Delta$$

it is a non-convex example with strong duality. The Lagrangian

$$L(d, \lambda) = f + \langle g, d \rangle + \frac{1}{2} \langle d | B + \lambda | d \rangle - \frac{\lambda}{2} \Delta^2$$

the dual function is

$$\inf_d L(d, \lambda) = \begin{cases} f - \langle g | (B + \lambda)^{-1} | g \rangle - \frac{\lambda}{2} \Delta^2 & B + \lambda \succeq 0 \\ -\infty & \text{else} \end{cases}$$

so the optimal value of the dual program is attained at a KKT point that satisfies

$$\begin{aligned} |d^*| \leq \Delta^2, \lambda^* \geq 0 \\ g + (B + \lambda^*) d^* = 0 \\ \lambda^* (|d^*|^2 - \Delta^2) = 0 \\ B + \lambda^* \succeq 0 \end{aligned}$$

such a point is a saddle point:

$$L(d^* + p, \lambda^*) = L(d^*, \lambda^*) + \frac{1}{2} \langle p|B + \lambda^*|p\rangle \geq L(d^*, \lambda^*)$$

and

$$L(d^*, \lambda^* + s) = L(d^*, \lambda^*) + \frac{s}{2}(|d^*|^2 - \Delta^2) \leq L(d^*, \lambda^*)$$

so the strong duality holds exactly at this KKT point.

So there are two cases, one is $\lambda^* = 0$ and d^* is just the Newton step. The other case is $|d^*| = \Delta$, and

$$d^* = -(B + \lambda^*)^{-1} g$$

If B has eigenvalues σ_i with increasing order, and $g = (g_i)$ under the canonical coordinates, we need to solve $\lambda + \sigma_1 \geq 0$ s.t.

$$\sum_i \frac{g_i^2}{(\sigma_i + \lambda)^2} = \Delta^2$$

if Δ is small we can get our λ^*.

In fact, strong duality holds for any programs with quadratic objective funcitons and exactly one quadratic constraints if Slater's condition holds (The proof of this is much more involved, seeA.4.1).

One can also use **conjugate gradient method** to solve the TRS.

2.3.2 Convergence analysis

2.4 Stochastic gradient descent

Chapter 3

First order optimization on manifolds

These are notes for chapter 4 and 11 of [1].

3.1 Riemannian gradient descent

Let M be a Riemanian manifold, and $f : M \to \mathbb{R}$ be a smooth function. Let R_x be the retraction at $x \in M$:i.e. $R_x : T_xM \to M$ is a smooth map s.t. $R_x(0) = x$ and $dR_x = id$. The Riemannian gradient descent (RGD) is defined as the iteration of the following update rule:

$$x_{k+1} = R_{x_k}(-t_k \nabla f(x_k))$$

where t_k is the step size.

3.1.1 Regularity and convergence

If f is bounded below, and if each step we have enough descent:

$$f(x_k) - f(x_{k+1}) \geqslant c|\nabla f(x_k)|^2 \tag{A}$$

Then

$$\min_{0 \leq k \leq N} |\nabla f(x_k)|^2 \leq \frac{f(x_0) - f(x_{N+1})}{c(N+1)} \leq \frac{f(x_0) - \inf f}{c(N+1)}$$

In particular the gradient will converge to zero, and any accumulation point of the sequence $\{x_k\}$ will be a critical point of f.

It's more natural to write the condition A as the Armijo-Goldstein condition

$$f(x) - f(R_x(-t\nabla f(x))) \geq rt|\nabla f(x)|^2 \tag{B}$$

where t is the step size, r is the ratio of the real descent and expected descent.

This can be done by control the "second order behavior" of f: if f is L-smooth w.r.t. the retraction, i.e.

$$f(R_x(s)) \leq f(x) + \langle \nabla f(x), s \rangle + \frac{L}{2}|s|^2 \tag{C}$$

for every $(x, s) \in TM$, then the descendence condition A holds if

$$t \in [t_{\min}, t_{\max}] \subset (0, 2/L)$$

In this case c can be chosen as

$$c = \inf \left\{ t - \frac{Lt^2}{2} \right\} = \min \left\{ t_{\max} - \frac{L}{2} t_{\max}^2, t_{\min} - \frac{L}{2} t_{\min}^2 \right\}$$

For example, if we choose $t_k = 1/L$, then $c = 1/(2L)$.

3.1.2 Backtrack line-search

In practice, we can not always find a fixed step size s.t. the condition A holds and we don't know a constant L beforehand. We can use backtrack line-search: we start with some initial step size t_0, and check if the condition B holds. If not, we shrink the step size by multiplying it with some $\tau \in (0, 1)$ until B is satisfied. The algorithm will terminate before

$$t \leq \frac{2(1-r)}{L}$$

after computing at most

$$\max \left(1, 2 + \log_\tau \left(\frac{2(1-r)}{tL} \right) \right)$$

retractions and evaluations as long as f satisfies the L-smoothness condition C.

Now we can choose some initial step size sequence t_k that $\sum_{k=0}^{\infty} t_k = \infty$, then at each step we have the descendence condition A with

$$c_k = r \min \left\{ t_k, \frac{2\tau(1-r)}{L} \right\}$$

As a consequence, we have

$$\min_{0 \leq k \leq K} |\nabla f(x_k)|^2 \leq \frac{f(x_0) - \inf f}{\sum_{k=0}^{K} c_k}$$

which implies $\lim_{k \to \infty} |\nabla f(x_k)| = 0$.

3.1.3 Local convergence

3.2 Convex optimization

In manifold case, we can define (strongly) convexity and Lipschitz smoothness in terms of geodesic segments: sets are convex if all geodesic segments connecting any two points in the set lie entirely

within the set, and functions are convex, strongly convex or Lipschitz smooth if their restriction to any geodesic is convex, strongly convex or Lipschitz smooth.

An alternative way to define μ-strong convexity is that $f(x) - f(y) - \mu\frac{d(x,y)^2}{2}$ being convex w.r.t. x. If x, y is joined by a minimizing geodesic with velocity $s \in T_yM$, let

$$D_f(x, y) = f(x) - f(y) - \langle \nabla f(y), s\rangle$$

we say f is μ-strongly convex if $D_f(x, y) \geq \frac{\mu}{2}d(x, y)^2$ where $d(\cdot, \cdot)$ is the Riemannian distance. This slightly different from the previous definition, but they are equivalent since the condition is essentially local: the function is (μ-strongly) convex if and only if it is (μ-strongly) convex at an open cover of S.

Appendix A

Appendix A: Linear algebra

A.1 Singular values

Consider an operator A, it will transform a sphere

$$\langle x, x \rangle = 1$$

to an ellipsoid

$$|Ax|^2 = \langle x, A^*Ax \rangle = \langle Ax, Ax \rangle$$

So we are looking at the quadratic form with matrice A^*A. The eigenvalue of A^*A is the square of the semi-axes of the ellipsoid. We thus listed the square root of A^*A in decreasing order

$$(\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r)$$

which are called the **singular values** of A.

The eigenvectors

$$A^*Av = \sigma^2 v$$

are called the right singular vectors, they form an unitary basis of the domain. The eigenvectors

$$AA^*u = \sigma^2 u$$

are called the left singular vectors which form a unitary basis for the codomain. Not hard to see that Av is a scalar multiple of a left singular vector, and A^*u is a scalar multiple of a right singular vector. Thus if we choose these vectors as bases, the matrix of A will be diagonal with singular values on the diagonal. As a consequence can always diagonalize A by orthogonal transformations on the domain and target, i.e. we have the following **SVD**

$$A = U\Sigma V^*$$

where U, V are unitary matrices and Σ is a rectangular diagonal matrix with singular values lying on the diagonal.

There is also a variational interpretation of singular values. Consider

$$(u, v) \mapsto \langle u, Av \rangle$$

on the products of unit sphere with constraints $|u|^2 = |v|^2 = 1$. Since the domain is compact we have at least one maximum. All extrema must satisfy the KKT condition

$$Av = 2\lambda_1 u$$
$$A^* u = 2\lambda_2 v$$

which implies $\sigma = 2\lambda_1 = 2\lambda$ and u, v are left and right singular vectors. Now we use induction by consider the unitary complement of u and v to find the rest of the singular values.

The **condition number** of A is the ratio of the largest and smallest singular value

$$\kappa = \frac{\sigma_1}{\sigma_r}$$

which can be viewed as a generalized eccentricity of the ellipsoid.

A.2 Sherman-Morrisan formula

Theorem A.2.1 (Sherman-Morrisan formula)

A is invertible, then $A + uv^*$ is invertible iff

$$1 + \langle v|A^{-1}|u\rangle \neq 0$$

in which case

$$(A + uv^*)^{-1} = A^{-1} - \frac{A^{-1}uv^*A^{-1}}{1 + \langle v|A^{-1}|u\rangle}$$

Proof. It suffices to prove the special case of $1 + uv^*$. Note $(uv^*)^k = \langle u, v\rangle^{k-1} uv^*$

$$(1 + uv^*)^{-1} = \sum_{k\geq 0}(-uv^*)^k = 1 - \frac{uv^*}{1 + \langle u, v\rangle}$$

This is just a formal computation, but the result indeed invert our $1+uv^*$. Necessarity is trivial. □

A.3 Gneralised inverse and Schur complement

Theorem A.3.1 (Rank-1 exactness)

If $A, B \in S$ are symmetric, then for every positive definite matrice $X \in S_+$, there is a vector x s.t.

$$\langle x|A|x\rangle = \mathrm{Tr}\,(AX)\,, \langle x|B|x\rangle = \mathrm{Tr}\,(BX)$$

As a consequence, the field of values of homogeneous quadratic forms

$$W(A,B) = \{(\langle x|A|x\rangle, \langle x|B|x\rangle)\}$$

is a convex cone.

TBA

A.4 S-procedure

Theorem A.4.1 (Strong duality for quadratic programs)

Strong duality holds for any programs with a quadratic objective funcitons and one quadratic constraints if Slater's condition holds.

The proof relies on the following lemma:

Lemma A.4.2 (S-procedure)

Supppose we have two quadratic forms

$$q_i(x) = \langle x|A_i|x\rangle + 2\langle b_i, x\rangle + c_i, i = 1, 2$$

If there exists $q_2(x_0) < 0$, then $q_2(x) \leq 0 \Rightarrow q_1(x) \geq 0$ iff there is $\lambda \geq 0$ s.t.

$$q_1 + \lambda q_2 \succeq 0$$

Proof. If part is trivial. To see the only if part, consider the closed cone generated by

$$\{(q_1(x), q_2(x))\}$$

The quadratic form have matrice

$$Q_i = \begin{pmatrix} A_i & b_i \\ b_i^* & c_i \end{pmatrix}$$

with

$$q_i(x) = \langle [x,1]|Q_j|[x,1]\rangle$$

According to A.3.1

$$q_i(x) + q_i(y) = \langle [z,t]|Q_i|[z,t]\rangle$$

if $t = 0$,

$$q_i(x + sz) = s^2 \langle z|A_i|z\rangle + 2s\langle A_i x + b_i|z\rangle + q_i(x)$$

which implies that $q_i(x)+q_i(y) = \langle z|A_i|z\rangle$ is in the closure of the cone, so the closed cone is convex. If there is a Slater point $(q_2(x_0)) < 0$, there must be a line

$$q_1 + \lambda q_2, \lambda \geq 0$$

that separates the closed cone and the third quadrant. □

TBA

Proof. Consider the problem

$$\min q_1 \text{ s.t. } q_2 \leq 0$$

The Lagrangian is

$$L(x, \lambda) = q_1(x) + \lambda q_2(x), \lambda \geq 0$$

the dual function is

$$\begin{aligned} -p^*(\lambda) &= \inf_x L(x, \lambda) \\ &= \begin{cases} c_1 + \lambda c_2 - \langle b_1 + \lambda b_2|(q_1 + \lambda A_2)^+|b_1 + \lambda b_2\rangle & q_1 + \lambda q_2 \succeq 0 \text{ and} \exists\ x \\ & \text{s.t. } (A_1 + \lambda A_2)x = b_1 + \lambda b_2 \\ -\infty & \text{else} \end{cases} \end{aligned}$$

□

Appendix B

Appendix B: Convex analysis

B.1 Strong convexity and Lipschitz smoothness

Throughout this section we assume f is continuously differentiable. The **Bregman divergence** or **Bregman distance** of f is defined as

$$D_f(x, y) = f(x) - f(y) - \langle \nabla f(y), x - y \rangle$$

It measures how much f deviates from its linear approximation at y. It's symmetrized version is

$$D_f(x, y) + D_f(y, x) = \langle \nabla f(x) - \nabla f(y), x - y \rangle$$

B.1.1 Definition (Strong convicity)

f is μ-convex if it satisfies one of the following equivalent conditions:

1. $f(x) - \frac{\mu}{2}\left|x\right|^2$ is convex.
2. (If f is differentiable)
$$tf(x) + (1-t)f(y) \geq f(tx + (1-t)y) + \frac{\mu}{2}t(1-t)\left|y - x\right|^2$$
for every x, y.
3. (If f is differentiable)
$$D_f(x, y) \geq \frac{\mu}{2}\left|y - x\right|^2$$
for every x, y.
4. (If f is differentiable)∇f is μ-strongly monotone
$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \mu\left|x - y\right|^2$$
for every x, y.
5. (If f is twice differentiable) $\nabla^2 f(x) \succeq \mu I$ for every x.

Proof. $4 \Rightarrow 1$: Note $D_f(-, y)$ has same convexity as f,

$$\nabla_x D_f(x, y) = \nabla f(x) - \nabla f(y)$$

□

Theorem B.1.2 (Properties of strong convexity)

For every x, y,

$$D_f(x, y) \leq \frac{1}{2\mu} |\nabla f(x) - \nabla f(y)|^2$$

and

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \leq \frac{1}{\mu} \left|\nabla f(x) - \nabla f(y)\right|^2$$

Proof. The twice Bregman divergence is

$$D_{D_f(-,y)}(y, x) = 0 - D_f(x, y) - \langle \nabla f(x) - \nabla f(y), y - x \rangle \geq \frac{\mu}{2} |y - x|^2$$

thus

$$D_f(x, y) \leq -\langle \nabla f(x) - \nabla f(y), y - x \rangle - \frac{\mu}{2} |y - x|^2 \leq \frac{1}{2\mu} |\nabla f(x) - \nabla f(y)|^2$$

The second inequality is a direct consequence of the first one. It can also be derived by Cauchy-Schwarz inequality and ∇f is Hölder. □

Similarly we have the following definition for Lipschitz smooth functions:

B.1.3 Definition (Lipschitz smoothness)

f is L-smooth if it satisfies one of the following equivalent conditions:

1. $f(x) - \frac{L}{2} \left|x\right|^2$ is concave.
2. (If f is differentiable)
$$tf(x) + (1 - t)f(y) \leq f(tx + (1 - t)y) + \frac{L}{2} t(1 - t) \left|y - x\right|^2$$
for every x, y.
3. (If f is differentiable)
$$D_f(x, y) \leq \frac{L}{2} \left|y - x\right|^2$$
for every x, y.
4. (If f is differentiable)
$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \leq L \left|x - y\right|^2$$
for every x, y.
5. (If f is twice differentiable) $\nabla^2 f(x) \preceq LI$ for every x.

If f is also convex, we also have the following counterpart of B.1.2:

Theorem B.1.4 (Bailon-Haddad)

f is convex and L-smooth is equivalent to the following:

1. For every x, y,
$$D_f(x,y) \geq \frac{1}{2L}|\nabla f(x) - \nabla f(y)|^2$$
2. f is **cocoercive**: for every x, y,
$$\langle \nabla f(x) - \nabla f(y), x - y\rangle \geq \frac{1}{L}\left|\nabla f(x) - \nabla f(y)\right|^2$$
3. f is convex and ∇f is L-Lipschitz.

Proof. As before $F_f(-,y)$ has same smoothness as f, so the twice Bregman divergence is
$$D_{D_f(-,y)}(z,x) = D_f(z,y) - D_f(x,y) - \langle \nabla f(x) - \nabla f(y), z - x\rangle \leq \frac{L}{2}|z-x|^2$$
The rest follows from Cauchy-Schwarz inequality. Convexity implies $D_f(z,y) \geq 0$, thus
$$D_f(x,y) \geq \frac{1}{2L}|\nabla f(x) - \nabla f(y)|^2$$
□

Now we consider strongly convex and Lipschitz smooth functions at the same time. We have the following important inequalities:

Theorem B.1.5 (Inequality for strong convex and Lipschitz smooth functions)

If f is μ-convex and L-smooth, for every x, y
$$D_f(x,y) \geq \frac{\mu L}{2(\mu+L)}\left|x-y\right|^2 + \frac{1}{2(\mu+L)}|\nabla f(x) - \nabla f(y)|^2$$
and
$$\langle \nabla f(x) - \nabla f(y), x - y\rangle \geq \frac{\mu L}{\mu+L}\left|x-y\right|^2 + \frac{1}{\mu+L}|\nabla f(x) - \nabla f(y)|^2$$

Proof. $f(x) - \frac{\mu}{2}\left|x\right|^2$ convex and $(L-\mu)$-smooth. Using cocoercivity in B.1.4 we get
$$D_f(x,y) - \frac{\mu}{2}|x-y|^2 \geq \frac{1}{2(L-\mu)}|\nabla f(x) - \nabla f(y) - \mu(x-y)|^2$$
This gives
$$LD_f(x,y) + \mu D_f(y,x) \geq \frac{\mu L}{2}|x-y|^2 + \frac{1}{2}|\nabla f(x) - \nabla f(y)|^2$$
Since right hand side is symmetric in x, y, we can exchange x, y to get
$$(L+\mu)D_f(y,x) \geq \frac{\mu L}{2}|x-y|^2 + \frac{1}{2}|\nabla f(x) - \nabla f(y)|^2$$
□

B.2 Subgradient and directional derivative

cyclic monotonicity

Bibliography

[1] Nicolas Boumal. An introduction to optimization on smooth manifolds. Cambridge University Press, 2023.

[2] Hu, J., Liu, X., Wen, ZW. et al. A Brief Introduction to Manifold Optimization. J. Oper. Res. Soc. China 8, 199-248 (2020).

[3] Dimitri P. Bertsekas. Convex Optimization Theory. Athena Scientific, 2009.

[4] Stephen Boyd and Lieven Vandenberghe. Convex Optimization. Cambridge University Press, 2004.

[5] Rockafellar, R. T. Convex Analysis. Princeton University Press, 1970.

[6] de Klerk, E., Glineur, F. & Taylor, A.B. On the worst-case complexity of the gradient method with exact line search for smooth strongly convex functions. Optim Lett 11, 1185-1199 (2017).

[7] Taylor, A. B., Hendrickx, J. M., & Glineur, F. (2017). Smooth strongly convex interpolation and exact worst-case performance of first-order methods. Mathematical Programming, 161(1-2), 307-345.

[8] 袁亮、黄红选、王晓军. 最优化：建模、算法与理论. 高教出版社, 2022.